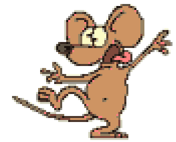
Mouse, the Language



URL: http://primepuzzle.com/mouse/mouse2.html

Last changed on Monday, April 23, 2007

Here's a program in Mouse which asks you to enter a number between 13 and 29 (inclusive) and then tells you what was entered.

```
~ num.mse
0 a:
(a. 12 > 30 a. > * 0 = ^
 "!Please enter a number between 13 and 29 (inclusive) : " ? a:)
"!You entered the number " a. ! "." $
```

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1. Comments start with the tilde (~).

2. There are 26 variables in Mouse. You may use lower case a thru z or upper case A thru Z. The assignment command character is the colon (:). The second line pushes a 0 onto Stack, then the address of the variable a is pushed onto Stack and then the colon pops the two operands off Stack and stores the value of the second at the address specified by the first. ie. a=0.

3. Loops are indicated by parentheses. When you reach the caret (^) the Stack is popped and, if zero or negative, control goes to the command after the trailing right parenthesis. If positive, the commands after the caret are run.

The period (.) is the "dereference" or "evaluate" command. It pops the operand (which will be a variable's address) off Stack and pushes the value at that address onto Stack. So a. means "replace the top of Stack with the value of a".

Numbers, like 12 and 30 and 0 etc. are multi-character commands and are pushed onto Stack. The comparison operators >, < and = pop two operands off Stack and push 0 (false) or 1 (true) onto Stack if the relationship is false or true. The arithmetic operators *, /, +, - etc. pop two operands off Stack, do the operation and then push the result onto Stack. Looking at line 3 above, the only way the Stack will have a positive value on it when you reach the caret is if a is not in the required range. Note the final 0 = which "flips" the condition. The condition, in "pseudo-code," is "not (a>12 and 30>a)."

This is worth walking thru in detail. Say a has the value 39, which is out of range. The a. 12 > will push a 1 on Stack (because a is indeed larger than 12). The 30 a. > will push a 0 on Stack (because 30 is not greater than 39). The multiply operator (*) will replace the 1

and the 0 with 0, because 1 times 0 is 0. The 0 = will replace the 0 with a 1, because 0 does equal 0. Since we're at the caret and Stack has a 1 on top (true), the loop continues and the user is asked to enter another number.

Welcome to Mouse!

4. ? is the read number from keyboard and push on Stack command.

5. The first ! in line 5 above causes a new line to display. The second ! is the pop number off Stack and display it command.

```
B0>mouse-p
MOUSE, Peter Grogono, 1983; Lee Bradley, 2007
Enter Mouse program's name : num.mse

Please enter a number between 13 and 29 (inclusive) : 36

Please enter a number between 13 and 29 (inclusive) : 17

You entered the number 17.
B0>
```